EXP 1:Write the python program to solve 8-Puzzle problem

```
from collections import deque

def solve(start, goal):
    queue = deque([(start, [])])
    visited = set()

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        if state in visited:
            continue

        visited.add(state)

        zero = state.index(0)
        row = zero // 3
        col = zero % 3

        moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]

        for dr, dc in moves:
            r = row + dr
            c = col + dc

            if 0 <= r < 3 and 0 <= c < 3:
                new_zero = r * 3 + c
                new_state = list(state)

                new_state[zero], new_state[new_zero] = \
                    new_state[new_zero], new_state[zero]

                new_state = tuple(new_state)
```

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            if new_state not in visited:
                queue.append((new_state, path + [state]))

    return None


start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

solution = solve(start, goal)

print("8-Puzzle Solution:")

for state in solution:
    print(state[0:3])
    print(state[3:6])
    print(state[6:9])
    print()
```

OUTPUT:

